



DR ESLAM AHMED
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Vectors

A vector is a journey: known roads only, against the arrow change the sign. Parallel = a number times.

THE TOPIC ON ONE MAP

Vectors

01 Column vectors and magnitude

$a + 2b$

$|\vec{HF}|$

find k

02 Vector paths

$\vec{AB} = b - a$

hexagons

03 Ratios on a line

$AP : PB = 1 : 3$

find λ, μ

04 Proof: parallel, straight line

$\vec{AB} = k\vec{CD}$

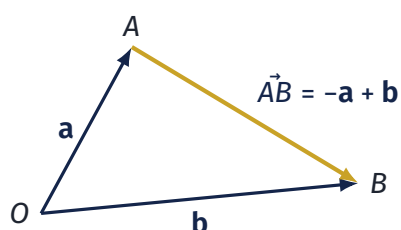
common point

05 Intersections

two routes to P

match a and b

THE JOURNEY Only use roads you know; against an arrow, change the sign:



$A \rightarrow O$: against $a \rightarrow -a$

$O \rightarrow B$: with $b \rightarrow +b$

01 Column Vectors and Magnitude

Add and subtract column vectors number by number. Magnitude is the length: Pythagoras on the top and bottom numbers.

WHEN YOU SEE → YOU DO

$a + 2b$ with column vectors

→ multiply, then add top with top, bottom with bottom

“magnitude of $\vec{HF}$ ”

→ build $\vec{HF}$ first, then $\sqrt{x^2 + y^2}$

“ $2m + n$ is parallel to $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$ ”

→ the two numbers in the same ratio

“find m and n ”

→ top equation and bottom equation, solve together

WORKED MODEL • NOVEMBER 2024 P2H

$\vec{FG} = \begin{pmatrix} -5 \\ 2 \end{pmatrix}$, $\vec{HG} = \begin{pmatrix} 4 \\ 14 \end{pmatrix}$. Find $|\vec{HF}|$.

1 $\vec{HF} = \vec{HG} + \vec{GF} = \begin{pmatrix} 4 \\ 14 \end{pmatrix} - \begin{pmatrix} -5 \\ 2 \end{pmatrix} = \begin{pmatrix} 9 \\ 12 \end{pmatrix}$

journey

ANSWER $\sqrt{9^2 + 12^2} = 15$

IDEA 01

Practise: Column Vectors and Magnitude

My questions first, then selected practice, then real past papers.

Q1 DR ESLAM Magnitude

CH 5 • TOPIC 1 2 marks

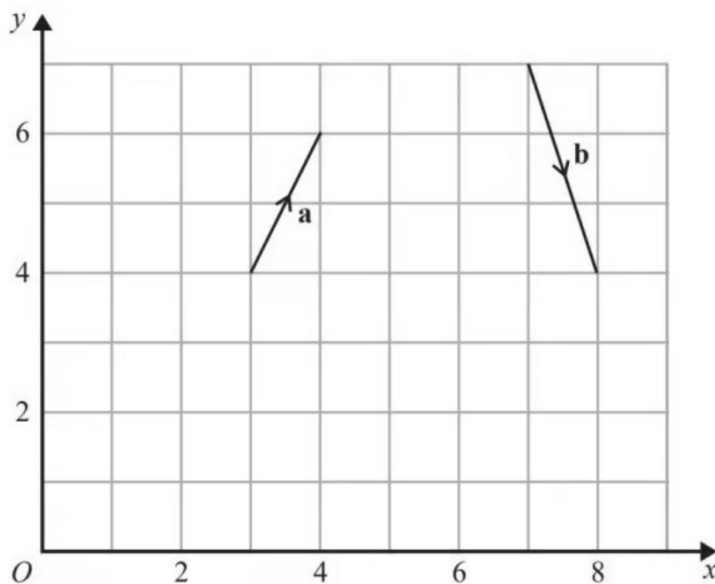
$\mathbf{a} = \begin{pmatrix} 3 \\ -4 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 1 \\ 6 \end{pmatrix}$. Find $2\mathbf{a} + \mathbf{b}$ and its magnitude.

SELECTED PRACTICE

Q2 PRACTICE Practice • Medium

CH 5 • TOPIC 1 3 marks

1 (a) The vector **a** and the vector **b** are shown on the grid.



On the grid, draw and label vector $-2\mathbf{a}$

(1 mark)

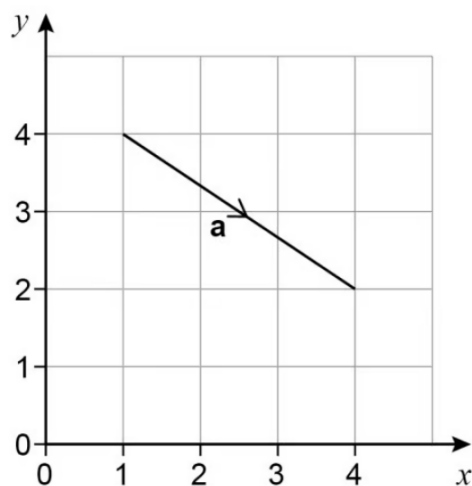
(b) Work out $\mathbf{a} + 2\mathbf{b}$ as a column vector.

(2 marks)

Q3**PRACTICE**

Practice • Medium

CH 5 • TOPIC 1

1 mark**13** Here is vector **a**.Circle the column vector that represents **a**.

$\begin{pmatrix} 3 \\ 2 \end{pmatrix}$	$\begin{pmatrix} -3 \\ 2 \end{pmatrix}$	$\begin{pmatrix} 3 \\ -2 \end{pmatrix}$	$\begin{pmatrix} -3 \\ -2 \end{pmatrix}$
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(1 mark)

Q4**PRACTICE**

Practice • Hard

CH 5 • TOPIC 1 5 marks

9 (a) The diagram shows parallelogram $ABCD$.

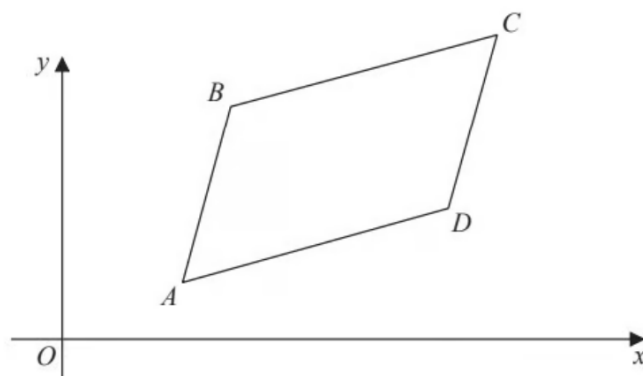


Diagram NOT
accurately drawn

$$\vec{AB} = \begin{pmatrix} 2 \\ 7 \end{pmatrix} \quad \vec{AC} = \begin{pmatrix} 10 \\ 11 \end{pmatrix}$$

The point B has coordinates $(5, 8)$

Work out the coordinates of the point C .

(3 marks)

(b) The point E has coordinates $(63, 211)$

Use a vector method to prove that ABE is a straight line.

(2 marks)

Q5**PRACTICE**

Practice • Very Hard

CH 5 • TOPIC 1 • 4 marks

14 Vector $\mathbf{m} = \begin{pmatrix} 2 \\ k \end{pmatrix}$ and vector $\mathbf{n} = \begin{pmatrix} 3 \\ 11 \end{pmatrix}$

Vector $2\mathbf{m} + \mathbf{n}$ is parallel to $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$

Find the value of k .

$$k = \dots\dots\dots$$

(4 marks)

PAST PAPERS

Q6 **PAST PAPER** Edexcel 4MA1 • January 2021 • Paper 1HR • Q13**CH 5 • TOPIC 1** **2 marks**

13 Here are two vectors.

$$\overrightarrow{AB} = \begin{pmatrix} 5 \\ 3 \end{pmatrix} \quad \overrightarrow{CB} = \begin{pmatrix} -2 \\ 4 \end{pmatrix}$$

Find, as a column vector, $\overrightarrow{AC}$

(Total for Question 13 is 2 marks)

Q7**PAST PAPER**

Edexcel 4MA1 • May 2023 • Paper 1HR • Q19

CH 5 • TOPIC 1 5 marks

- 19 The diagram shows a triangle ABC where A , B and C represent the positions of three towns.

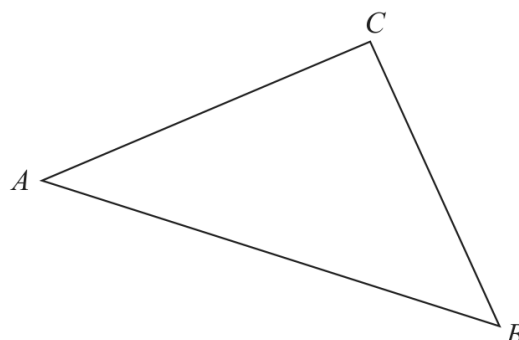


Diagram **NOT**
accurately drawn

$$\vec{AB} = \begin{pmatrix} 7 \\ -2 \end{pmatrix} \quad \vec{BC} = \begin{pmatrix} -3 \\ 5 \end{pmatrix}$$

Pru travels directly from A to B and then directly from B to C

Yang travels directly from A to C

Given that the values for $\vec{AB}$ and $\vec{BC}$ are in kilometres,

work out how much further Pru travels than Yang travels.

Give your answer in km, correct to one decimal place.

..... km

(Total for Question 19 is 5 marks)

Q8**PAST PAPER**

Edexcel 4MA1 • November 2024 • Paper 2H • Q17

CH 5 • TOPIC 1 3 marks

17 Here are two vectors.

$$\vec{FG} = \begin{pmatrix} -5 \\ 2 \end{pmatrix} \quad \vec{HG} = \begin{pmatrix} 4 \\ 14 \end{pmatrix}$$

Calculate the magnitude of the vector $\vec{HF}$

.....
(Total for Question 17 is 3 marks)

02 Vector Paths in Diagrams

Pick a route through points you know. Every arrow you go against changes sign. Simplify at the end.

WHEN YOU SEE → YOU DO

“find $\vec{AB}$ ”, O at the start

$$\longrightarrow \vec{AO} + \vec{OB} = -\mathbf{a} + \mathbf{b}$$

a parallelogram

$\longrightarrow$ opposite sides are the same vector

a regular hexagon with centre O

$$\longrightarrow \vec{OC} = \mathbf{b} - \mathbf{a}; \text{ opposite sides equal}$$

“midpoint”

$\longrightarrow$ half the vector

WORKED MODEL • JUNE 2025 P2HR

$\vec{OA} = 4\mathbf{a}$, $\vec{OB} = 4\mathbf{b}$, $AP : PB = 1 : 3$. Find $\vec{OP}$.

1 $\vec{AB} = 4\mathbf{b} - 4\mathbf{a}$

journey $A \rightarrow O \rightarrow B$

2 $\vec{OP} = 4\mathbf{a} + \frac{1}{4}(4\mathbf{b} - 4\mathbf{a})$

$\frac{1}{4}$ of the way

ANSWER $3\mathbf{a} + \mathbf{b}$

IDEA 02

Practise: Vector Paths in Diagrams

My questions first, then selected practice, then real past papers.

Q9

DR ESLAM Path

CH 5 • TOPIC 1 2 marks

$\vec{OA} = \mathbf{a}$, $\vec{OB} = \mathbf{b}$. Find $\vec{AB}$ and $\vec{BA}$.

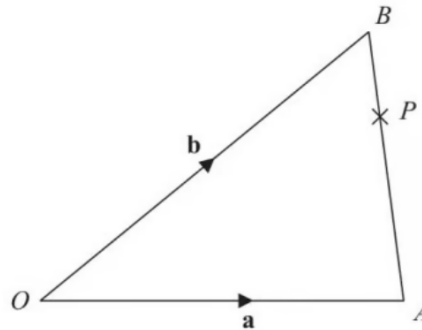
SELECTED PRACTICE

Q10**PRACTICE**

Practice • Medium

CH 5 • TOPIC 1 4 marks

2 (a)

Diagram **NOT**
accurately drawn OAB is a triangle.

$$\vec{OA} = \mathbf{a}$$

$$\vec{OB} = \mathbf{b}$$

Find $\vec{AB}$ in terms of $\mathbf{a}$ and $\mathbf{b}$.

(1 mark)

(b) P is the point on AB such that $AP : PB = 3 : 1$ Find $\vec{OP}$ in terms of $\mathbf{a}$ and $\mathbf{b}$.

Give your answer in its simplest form.

(3 marks)

Q11**PRACTICE**

Practice • Medium

CH 5 • TOPIC 1 4 marks

8 $ABCDEF$ is a regular hexagon.

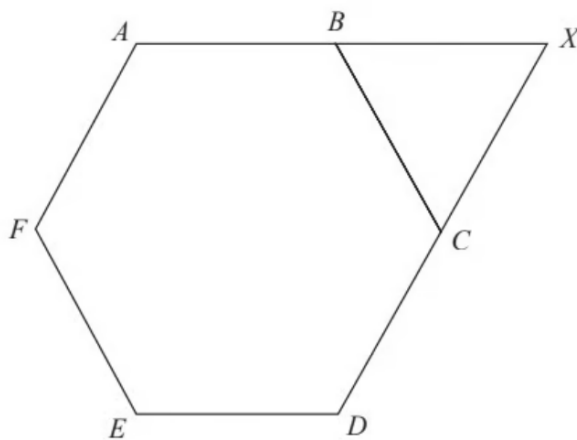


Diagram **NOT**
accurately drawn

ABX and DCX are straight lines.

$$\overrightarrow{AB} = \mathbf{a}$$

$$\overrightarrow{BC} = \mathbf{b}$$

Find $\overrightarrow{EX}$ in terms of $\mathbf{a}$ and $\mathbf{b}$.

Give your answer in its simplest form.

(4 marks)

Q12**PRACTICE**

Practice • Hard

CH 5 • TOPIC 1 1 mark

12 PQR is a straight line.

$$PQ : QR = 3 : 1$$

$$\vec{PQ} = \mathbf{a}$$

Not drawn
accuratelyCircle the vector $\vec{RQ}$

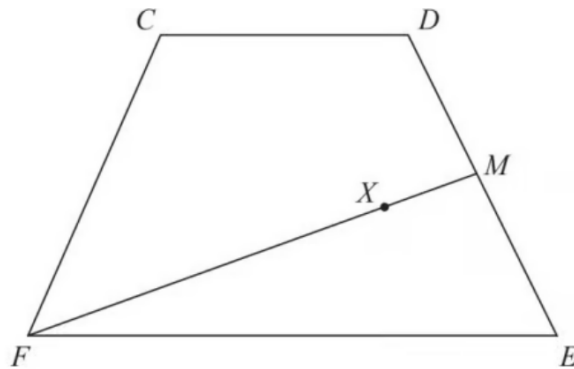
$\frac{1}{3}\mathbf{a}$	$\frac{1}{4}\mathbf{a}$	$-\frac{1}{3}\mathbf{a}$	$-\frac{1}{4}\mathbf{a}$
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(1 mark)

Q13**PRACTICE**

Practice · Very Hard

CH 5 • TOPIC 1

6 marks7 (a) $CDEF$ is a quadrilateral.

$$\overrightarrow{CD} = \mathbf{a}, \quad \overrightarrow{DE} = \mathbf{b} \quad \text{and} \quad \overrightarrow{FC} = \mathbf{a} - \mathbf{b}.$$

Express $\overrightarrow{FE}$ in terms of $\mathbf{a}$ and/or $\mathbf{b}$.
Give your answer in its simplest form.

(2 marks)

- (b) M is the midpoint of DE .
 X is the point on FM such that $FX:XM = n:1$
 CXE is a straight line.

Work out the value of n .

(4 marks)

PAST PAPERS

Q14 PAST PAPER Edexcel 4MA1 • November 2021 • Paper 1H • Q14

CH 5 • TOPIC 1 8 marks

14 $ABCDEF$ and $GHIJKL$ are regular hexagons each with centre O .

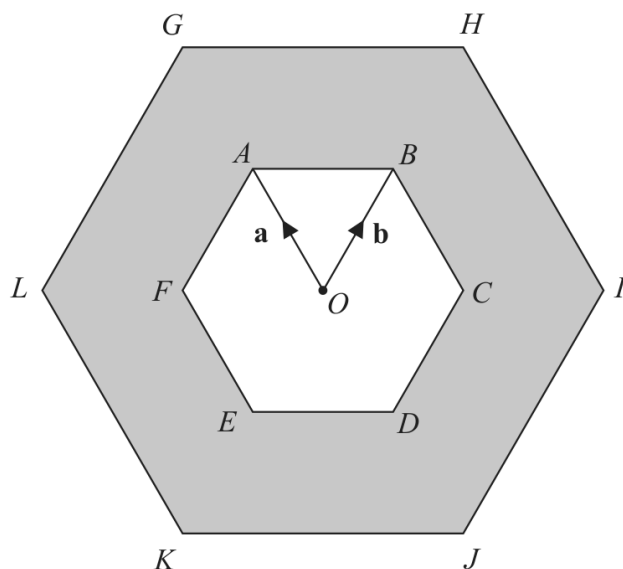


Diagram **NOT**
accurately drawn

$GHIJKL$ is an enlargement of $ABCDEF$, with centre O and scale factor 2

$$\vec{OA} = \mathbf{a} \quad \vec{OB} = \mathbf{b}$$

(a) Write the following vectors, in terms of $\mathbf{a}$ and $\mathbf{b}$.
Simplify your answers.

(i) $\vec{AB}$

.....
(1)

(ii) $\vec{KI}$

.....
(2)

(iii) $\vec{LD}$

.....

The triangle OAB has an area of 5 cm^2

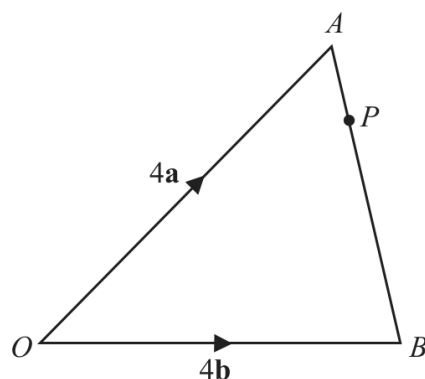
(b) Calculate the area of the shaded region.

..... cm^2

(Total for Question 14 is 8 marks)

Q15**PAST PAPER**

Edexcel 4MA1 • June 2025 • Paper 2HR • Q18

CH 5 • TOPIC 1 3 marks18 OAB is a triangle.Diagram **NOT**
accurately drawn

$$\vec{OA} = 4\mathbf{a}$$

$$\vec{OB} = 4\mathbf{b}$$

P is the point on AB such that $AP : PB = 1 : 3$

(a) Write down $\vec{AB}$ in terms of $\mathbf{a}$ and $\mathbf{b}$

.....

(b) Express $\vec{OP}$ in terms of $\mathbf{a}$ and $\mathbf{b}$
Give your answer in its simplest form.

.....

(Total for Question 18 is 3 marks)

03 Ratios on a Line

$AP : PB = m : n$ means P is $\frac{m}{m+n}$ of the way from A to B . Always go along the whole line $\vec{AB}$ first.

WHEN YOU SEE → YOU DO

“ $AP : PB = 5 : 3$ ”

→ $\vec{AP} = \frac{5}{8}\vec{AB}$

“ $AB : BR = 1 : 2$ ”

→ $\vec{BR} = 2\vec{AB}$

“given $\vec{OP} = k(3\mathbf{a} + 5\mathbf{b})$ ”

→ find $\vec{OP}$, then compare

an unknown ratio $\lambda : 1$

→ write the point with λ , use a straight-line fact

WORKED MODEL • RATIO

$\vec{OA} = 2\mathbf{a}$, $\vec{OB} = 2\mathbf{b}$, $AP : PB = 5 : 3$, $\vec{OP} = k(3\mathbf{a} + 5\mathbf{b})$. Find k .

1 $\vec{OP} = 2\mathbf{a} + \frac{5}{8}(2\mathbf{b} - 2\mathbf{a}) = \frac{3}{4}\mathbf{a} + \frac{5}{4}\mathbf{b}$

$\frac{5}{8}$ of AB

ANSWER $\frac{1}{4}(3\mathbf{a} + 5\mathbf{b}) \Rightarrow k = \frac{1}{4}$

IDEA 03

Practise: Ratios on a Line

My questions first, then selected practice, then real past papers.

Q16 DR ESLAM Ratio

CH 5 • TOPIC 1 3 marks

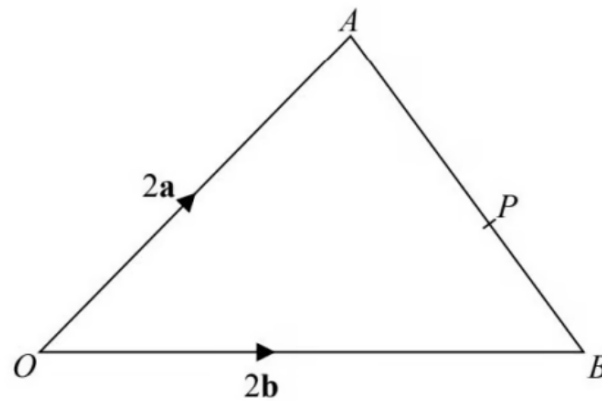
$\vec{OA} = 6\mathbf{a}$, $\vec{OB} = 6\mathbf{b}$. P lies on AB with $AP : PB = 1 : 2$. Find $\vec{OP}$.

SELECTED PRACTICE

Q17 PRACTICE Practice • Hard

CH 5 • TOPIC 1 4 marks

6



OAB is a triangle. P is the point on AB such that $AP:PB = 5:3$

$$\vec{OA} = 2\mathbf{a}$$

$$\vec{OB} = 2\mathbf{b}$$

$$\vec{OP} = k(3\mathbf{a} + 5\mathbf{b}) \text{ where } k \text{ is a scalar quantity.}$$

Find the value of k .

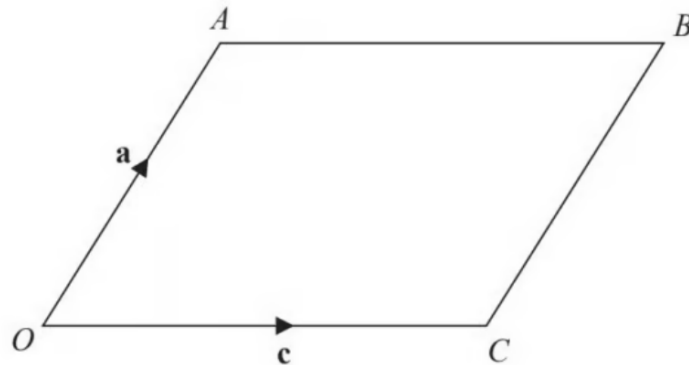
(4 marks)

Q18**PRACTICE**

Practice • Very Hard

CH 5 • TOPIC 1 4 marks

4



$OABC$ is a parallelogram.

$\vec{OA} = \mathbf{a}$ and $\vec{OC} = \mathbf{c}$

X is the midpoint of the line AC .

OCD is a straight line so that $OC : CD = k : 1$

Given that $\vec{XD} = 3\mathbf{c} - \frac{1}{2}\mathbf{a}$

find the value of k .

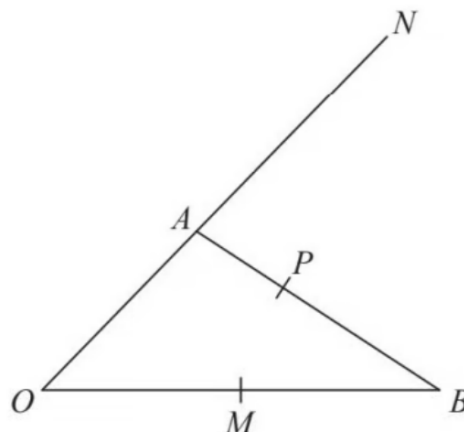
(4 marks)

Q19**PRACTICE**

Practice • Very Hard

CH 5 • TOPIC 1 5 marks

5



OAN , OMB and APB are straight lines.

$AN = 2OA$.

M is the midpoint of OB .

$$\overrightarrow{OA} = \mathbf{a} \quad \overrightarrow{OB} = \mathbf{b}$$

$$\overrightarrow{AP} = k\overrightarrow{AB} \text{ where } k \text{ is a scalar quantity.}$$

Given that MPN is a straight line, find the value of k .

(5 marks)

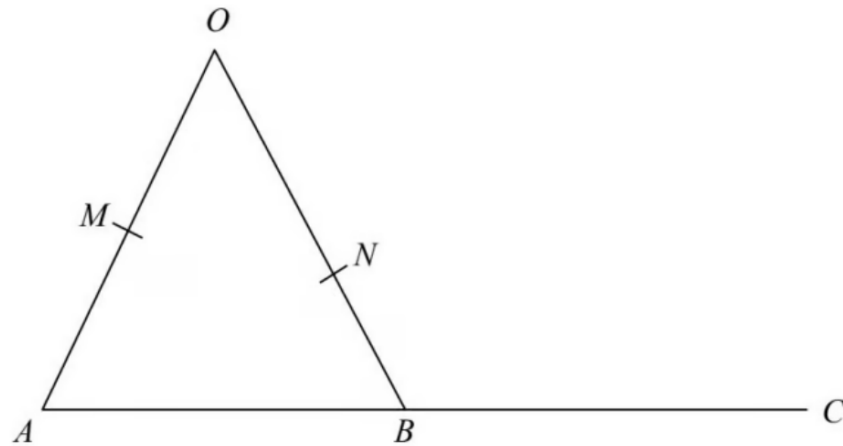
Q20**PRACTICE**

Practice • Very Hard

CH 5 • TOPIC 1

5 marks

6



OMA , ONB and ABC are straight lines.

M is the midpoint of OA .

B is the midpoint of AC .

$\vec{OA} = 6\mathbf{a}$ $\vec{OB} = 6\mathbf{b}$ $\vec{ON} = k\mathbf{b}$ where k is a scalar quantity.

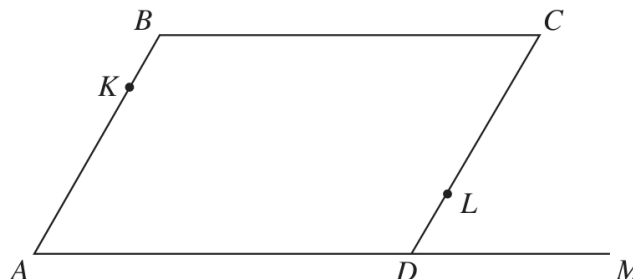
Given that MNC is a straight line, find the value of k .

(5 marks)

PAST PAPERS

Q21 PAST PAPER Edexcel 4MA1 • June 2022 • Paper 2HR • Q25

CH 5 • TOPIC 1 5 marks

25 $ABCD$ is a parallelogram and ADM is a straight line.Diagram **NOT**
accurately drawn

$$\vec{AB} = \mathbf{a} \quad \vec{BC} = \mathbf{b} \quad \vec{DM} = \frac{1}{2} \mathbf{b}$$

 K is the point on AB such that $AK:AB = \lambda:1$ L is the point on CD such that $CL:CD = \mu:1$ KLM is a straight line.Given that $\lambda:\mu = 1:2$ use a vector method to find the value of λ and the value of μ

$$\lambda = \dots\dots\dots$$

$$\mu = \dots\dots\dots$$

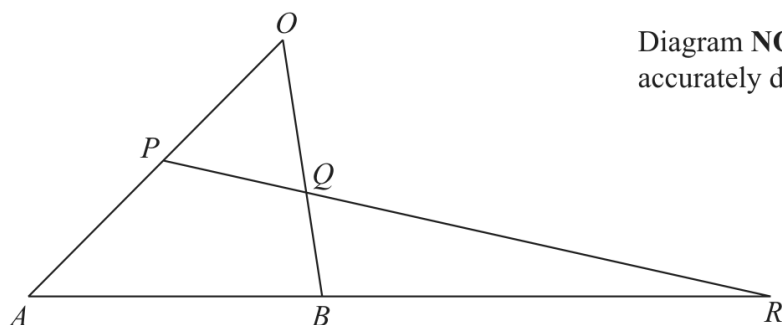
(Total for Question 25 is 5 marks)

Q22**PAST PAPER**

Edexcel 4MA1 • June 2025 • Paper 1H • Q23

CH 5 • TOPIC 1 5 marks

23

Diagram **NOT**
accurately drawn OAB is a triangle. P is the midpoint of OA Q is a point on OB ABR and PQR are straight lines.

$$\vec{OA} = 12\mathbf{a} \quad \vec{OB} = 8\mathbf{b}$$

(a) Express $\vec{AB}$ in terms of $\mathbf{a}$ and $\mathbf{b}$

.....

$$AB:BR = 1:2 \quad \vec{OQ} = n\mathbf{b}$$

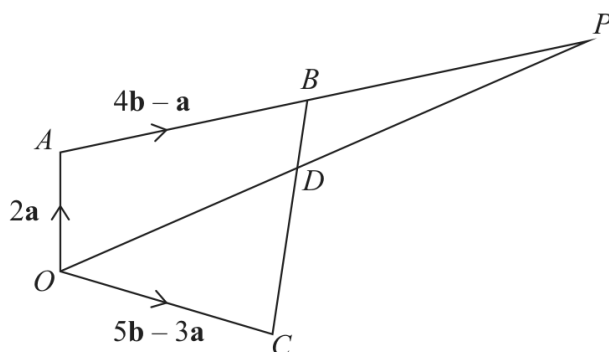
(b) Use a vector method to find the value of n $n =$ **(Total for Question 23 is 5 marks)**

Q23**PAST PAPER**

Edexcel 4MA1 • November 2025 • Paper 1H • Q27

CH 5 • TOPIC 1 6 marks

27

Diagram **NOT**
accurately drawn

$OABC$ is a quadrilateral.
 ABP and ODP are straight lines.

$$\vec{OA} = 2\mathbf{a} \quad \vec{AB} = 4\mathbf{b} - \mathbf{a} \quad \vec{OC} = 5\mathbf{b} - 3\mathbf{a}$$

- (a) Find an expression in terms of $\mathbf{a}$ and $\mathbf{b}$ for the vector $\vec{BC}$
 Simplify your answer.

The point D lies on BC such that $BD:DC = 1:3$

Given that $\vec{OP} = n\vec{OD}$

- (b) use a vector method to find the value of n

$n = \dots\dots\dots$

(Total for Question 27 is 6 marks)

04 Proof: Parallel and Straight Lines

Parallel: show one vector is a number times the other. Straight line: parallel and a common point. Finish with the sentence.

WHEN YOU SEE → YOU DO

“show that AB is parallel to MN ”

$$\rightarrow \vec{AB} = k\vec{MN}$$

“show that AGN is a straight line”

$$\rightarrow \vec{AN} = k\vec{AG}, \text{ common point } A$$

“is BCD a straight line?”

$$\rightarrow \text{compare } \vec{BC} \text{ and } \vec{CD}; \text{ no multiple} \rightarrow \text{No}$$

“prove $ABDC$ is a trapezium”

$$\rightarrow \text{one pair of sides parallel}$$

ZONE PARALLEL

$$\vec{PQ} = k\vec{RS}$$

same direction, k times as long

ZONE STRAIGHT LINE

parallel + a common point

$$\vec{AB} = k\vec{AC}$$

ZONE RATIO $m : n$

$$\vec{AP} = \frac{m}{m+n}\vec{AB}$$

fraction of the whole line

WORKED MODEL • STRAIGHT LINE

$\vec{OA} = 6\mathbf{a}$, $\vec{OB} = 6\mathbf{b}$, M mid AB , N mid OB , $OG : GM = 2 : 1$. Show AGN is straight.

1 $\vec{OG} = \frac{2}{3}(3\mathbf{a} + 3\mathbf{b}) = 2\mathbf{a} + 2\mathbf{b}$

ratio

2 $\vec{AG} = -4\mathbf{a} + 2\mathbf{b}$, $\vec{AN} = -6\mathbf{a} + 3\mathbf{b}$

journeys

ANSWER $\vec{AN} = \frac{3}{2}\vec{AG}$ and A is common: straight line

IDEA 04

Practise: Proof: Parallel and Collinear

My questions first, then selected practice.

Q24 DR ESLAM Parallel

CH 5 • TOPIC 1 2 marks

Show that $6\mathbf{a} - 9\mathbf{b}$ is parallel to $2\mathbf{a} - 3\mathbf{b}$.

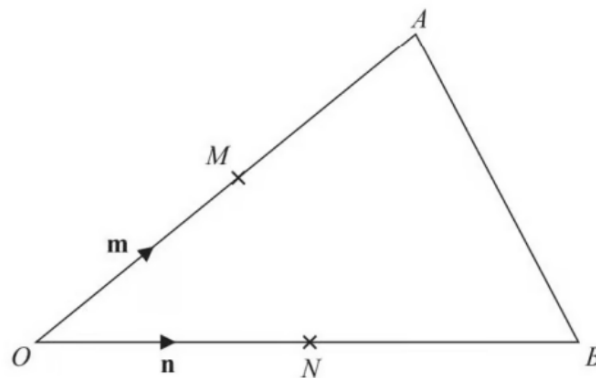
SELECTED PRACTICE

Q25**PRACTICE**

Practice • Medium

CH 5 • TOPIC 1 3 marks

4

Diagram **NOT**
accurately drawn OAB is a triangle. M is the midpoint of OA . N is the midpoint of OB .

$$\overrightarrow{OM} = \mathbf{m}$$

$$\overrightarrow{ON} = \mathbf{n}$$

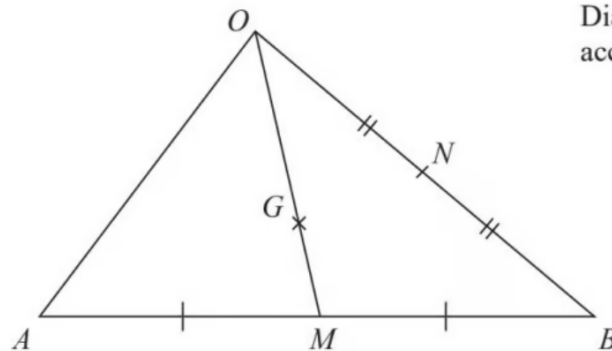
Show that AB is parallel to MN .

(3 marks)

Q26**PRACTICE**

Practice • Hard

CH 5 • TOPIC 1

6 marks**4 (a)**Diagram **NOT**
accurately drawn

$$\overrightarrow{OA} = 6\mathbf{a} \text{ and } \overrightarrow{OB} = 6\mathbf{b}$$

M is the midpoint of AB .

Write $\overrightarrow{OM}$ in terms of $\mathbf{a}$ and $\mathbf{b}$.

Give your answer in its simplest form.

(2 marks)

(b) N is the midpoint of OB .

G is the point on OM such that $OG : GM = 2 : 1$

Show that AGN is a straight line.

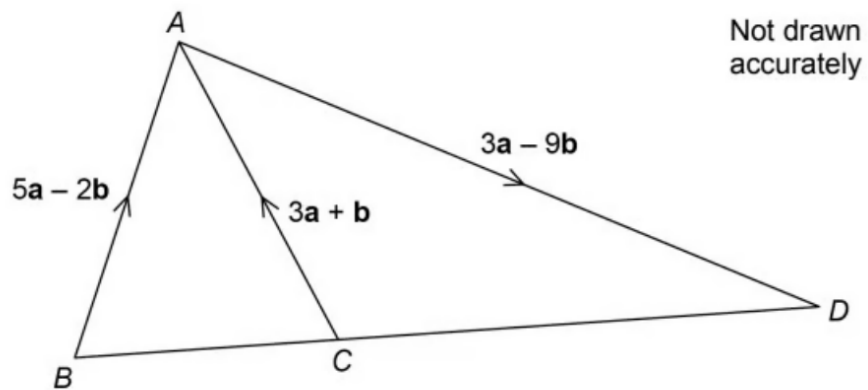
(4 marks)

Q27**PRACTICE**

Practice · Hard

CH 5 · TOPIC 1 3 marks

14



Is BCD a straight line?

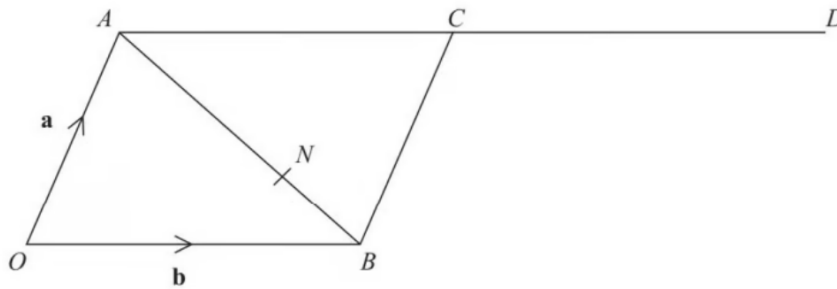
Show working to support your answer.

(3 marks)

Q28**PRACTICE**

Practice • Very Hard

CH 5 • TOPIC 1

6 marks1 (a) $OACB$ is a parallelogram.Diagram NOT
accurately drawn

$$\vec{OA} = \mathbf{a} \text{ and } \vec{OB} = \mathbf{b}$$

D is the point such that $\vec{AC} = \vec{CD}$

The point N divides AB in the ratio 2: 1

Write an expression for $\vec{ON}$ in terms of $\mathbf{a}$ and $\mathbf{b}$.

(3 marks)

(b) Prove that OND is a straight line.

(3 marks)

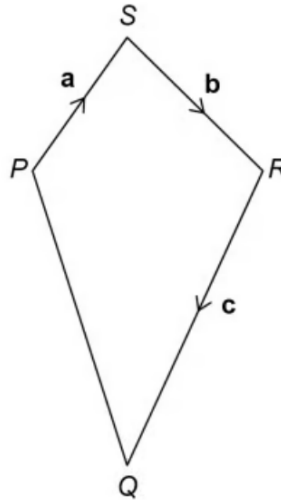
Q29**PRACTICE**

Practice • Very Hard

CH 5 • TOPIC 1 3 marks

12 Here is quadrilateral $PQRS$.

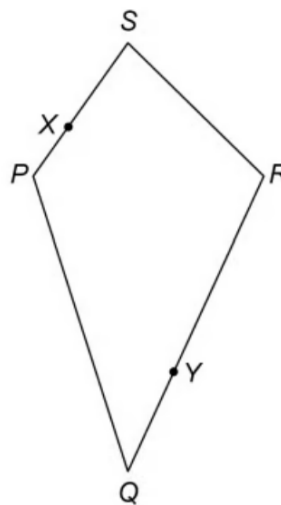
$$\vec{PS} = \mathbf{a} \quad \vec{SR} = \mathbf{b} \quad \vec{RQ} = \mathbf{c}$$



Not drawn
accurately

X is a point on PS where $PX : XS = 1 : 2$

Y is a point on RQ where $RY : YQ = 2 : 1$



Not drawn
accurately

Is XY parallel to PQ

Show working to support your answer.

(3 marks)

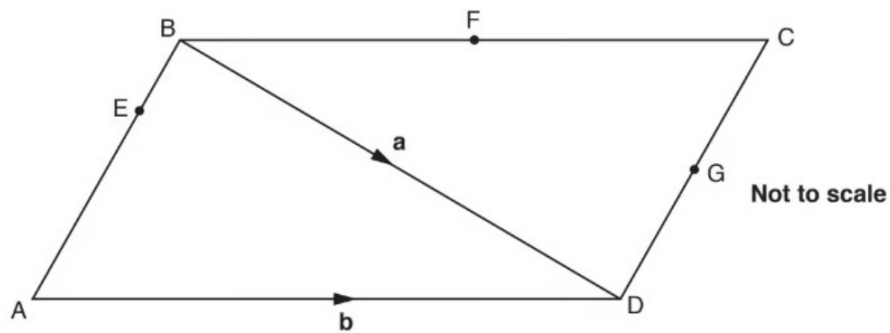
Q30**PRACTICE**

Practice • Very Hard

CH 5 • TOPIC 1

3 marks

16 ABCD is a parallelogram.



$$\vec{BD} = \mathbf{a} \text{ and } \vec{AD} = \mathbf{b}.$$

F is the midpoint of BC.

G is the midpoint of DC.

$$AE = 3EB.$$

Prove that $\vec{EF}$ and $\vec{AG}$ are parallel.

(3 marks)

05 Intersections: Two Routes to One Point

Dr Eslam's method: reach the crossing point P by two different routes, each with its own unknown (s, t). Set them equal and match $\mathbf{a}$ and $\mathbf{b}$.

WHEN YOU SEE → YOU DO

" P is the point of intersection of ON and AM "

$$\rightarrow \vec{OP} = s\vec{ON} \text{ and } \vec{OP} = \vec{OA} + t\vec{AM}$$

"find the ratio $AP : PB$ "

$$\rightarrow t \text{ gives the ratio: } t : (1 - t)$$

"use a vector method"

$\rightarrow$ the two-route equation must be shown

" $OY = k \times ON$ "

$\rightarrow k$ is one of your two unknowns

Route 1 with s

Route 2 with t

Equate

Match $\mathbf{a}$ and $\mathbf{b}$

Solve

WORKED MODEL • JANUARY 2022 P1H

$\vec{OA} = 8\mathbf{a}$, $\vec{OB} = 6\mathbf{b}$, $OM : MB = 1 : 2$, N mid AB , $P = ON \cap AM$. Find $\vec{OP}$.

1 route 1: $s(4\mathbf{a} + 3\mathbf{b})$; route 2: $8\mathbf{a} + t(2\mathbf{b} - 8\mathbf{a})$

two routes

2 $4s = 8 - 8t$, $3s = 2t \Rightarrow s = \frac{1}{2}$

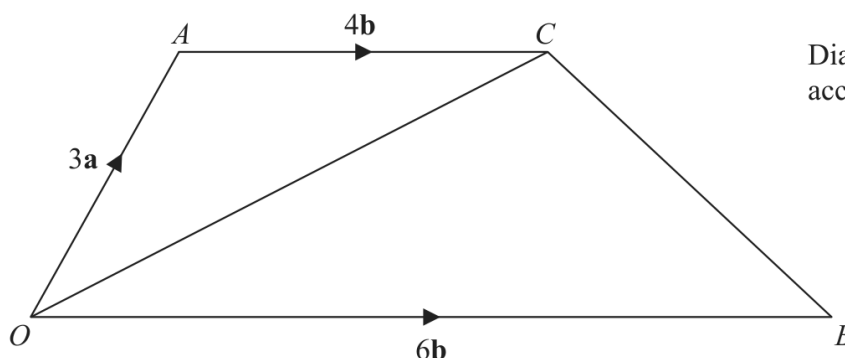
match $\mathbf{a}, \mathbf{b}$

ANSWER $\vec{OP} = 2\mathbf{a} + \frac{3}{2}\mathbf{b}$

IDEA 05

Practise: Intersections: Two Routes to One Point

My questions first, then real past papers.

PAST PAPERS**Q31** **PAST PAPER** Edexcel 4MA1 • January 2020 • Paper 2HR • Q26**CH 5 • TOPIC 1 5 marks****26** The diagram shows trapezium $OACB$.Diagram **NOT**
accurately drawn

$$\vec{OA} = 3\mathbf{a} \quad \vec{OB} = 6\mathbf{b} \quad \vec{AC} = 4\mathbf{b}$$

 N is the point on OC such that ANB is a straight line.Find $\vec{ON}$ as a simplified expression in terms of $\mathbf{a}$ and $\mathbf{b}$.

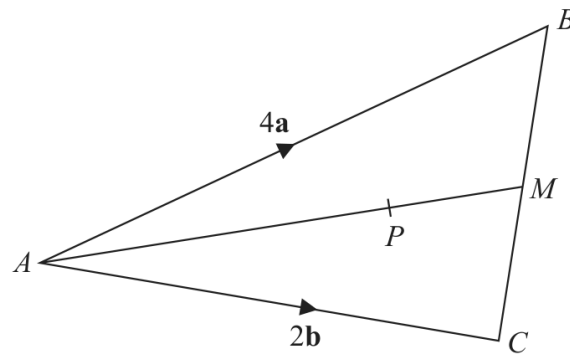
(Total for Question 26 is 5 marks)

Q32**PAST PAPER**

Edexcel 4MA1 • January 2020 • Paper 2H • Q23

CH 5 • TOPIC 1 3 marks

23

Diagram **NOT**
accurately drawn ABC is a triangle.The midpoint of BC is M . P is a point on AM .

$$\vec{AB} = 4\mathbf{a}$$

$$\vec{AC} = 2\mathbf{b}$$

$$\vec{AP} = \frac{3}{2}\mathbf{a} + \frac{3}{4}\mathbf{b}$$

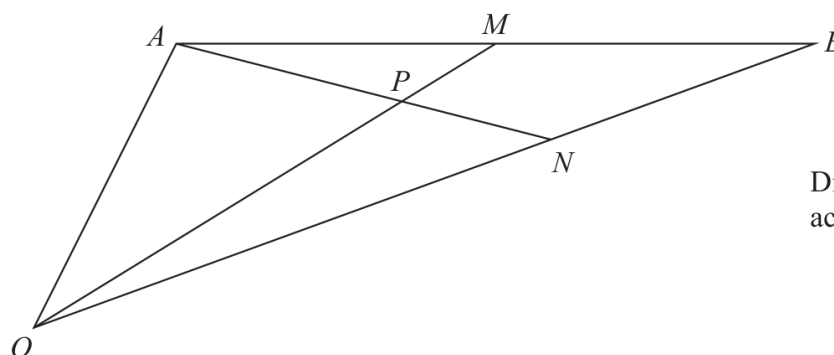
Find the ratio $AP:PM$

.....

(Total for Question 23 is 3 marks)

Q33**PAST PAPER**

Edexcel 4MA1 • January 2021 • Paper 1H • Q23

CH 5 • TOPIC 1 6 marks**23** OAB is a triangle.Diagram **NOT**
accurately drawn

$$\vec{OA} = 2\mathbf{a} \text{ and } \vec{OB} = 2\mathbf{b}$$

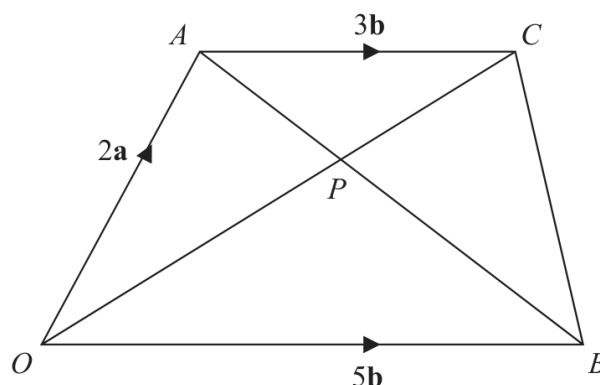
 M is the midpoint of AB . N is the point on OB such that $ON:NB = 2:1$ P is the point on AN such that OPM is a straight line.Use a vector method to find $OP:PM$

Show your working clearly.

(Total for Question 23 is 6 marks)

Q34 PAST PAPER Edexcel 4MA1 • May 2021 • Paper 1H • Q26

CH 5 • TOPIC 1 5 marks

26 $OACB$ is a trapezium.Diagram **NOT**
accurately drawn

$$\vec{OA} = 2\mathbf{a} \quad \vec{OB} = 5\mathbf{b} \quad \vec{AC} = 3\mathbf{b}$$

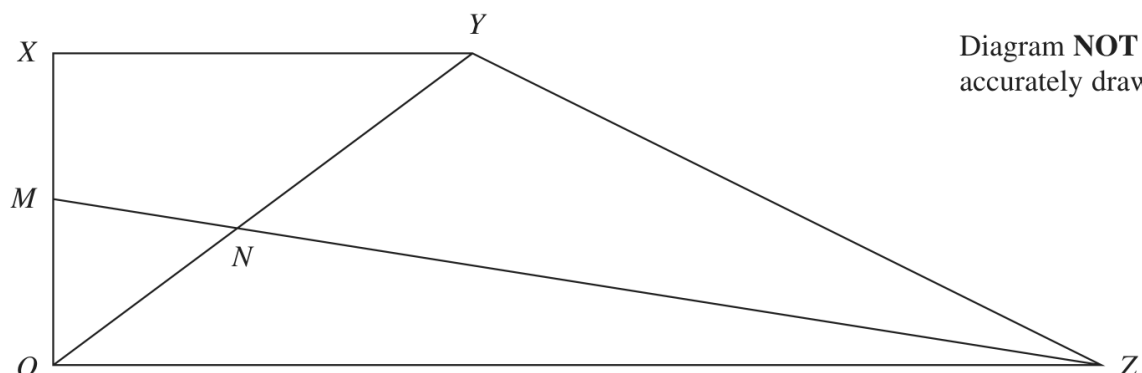
The diagonals, OC and AB , of the trapezium intersect at the point P .Find and simplify an expression, in terms of $\mathbf{a}$ and $\mathbf{b}$, for $\vec{OP}$
Show your working clearly.

$$\vec{OP} = \dots\dots\dots$$

(Total for Question 26 is 5 marks)

Q35 PAST PAPER Edexcel 4MA1 • January 2022 • Paper 1HR • Q24

CH 5 • TOPIC 1 5 marks

24 $OXYZ$ is a trapezium.

$$\overrightarrow{OX} = \mathbf{a}$$

$$\overrightarrow{XY} = \mathbf{b}$$

$$\overrightarrow{OZ} = 3\mathbf{b}$$

M is the midpoint of OX

N is the point such that MNZ and ONY are straight lines.

Given that $ON : OY = \lambda : 1$

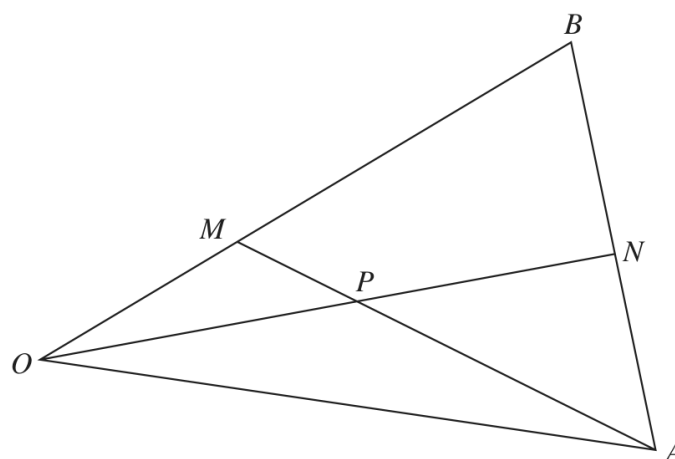
use a vector method to find the value of λ

$$\lambda = \dots\dots\dots$$

(Total for Question 24 is 5 marks)

Q36 PAST PAPER Edexcel 4MA1 • January 2022 • Paper 1H • Q22

CH 5 • TOPIC 1 5 marks

22 The diagram shows triangle OAB Diagram **NOT**
accurately drawn

$$\vec{OA} = 8\mathbf{a} \quad \vec{OB} = 6\mathbf{b}$$

 M is the point on OB such that $OM:MB = 1:2$ N is the midpoint of AB P is the point of intersection of ON and AM

Using a vector method, find $\vec{OP}$ as a simplified expression in terms of $\mathbf{a}$ and $\mathbf{b}$
 Show your working clearly.

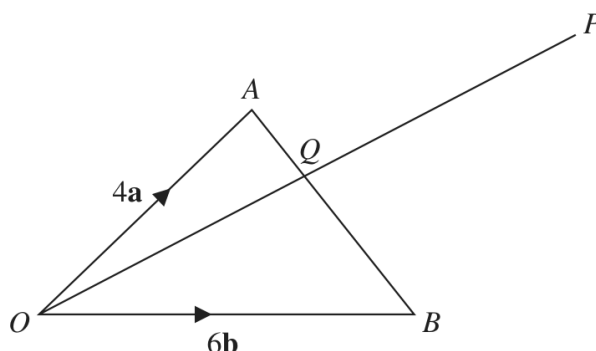
$$\vec{OP} = \dots\dots\dots$$

(Total for Question 22 is 5 marks)

Q37 PAST PAPER Edexcel 4MA1 • June 2022 • Paper 2H • Q21

CH 5 • TOPIC 1 5 marks

21

Diagram **NOT**
accurately drawn OAB is a triangle. Q is the point on AB such that OQP is a straight line.

$$\vec{OA} = 4\mathbf{a} \quad \vec{OB} = 6\mathbf{b} \quad \vec{AP} = 2\mathbf{a} + 8\mathbf{b}$$

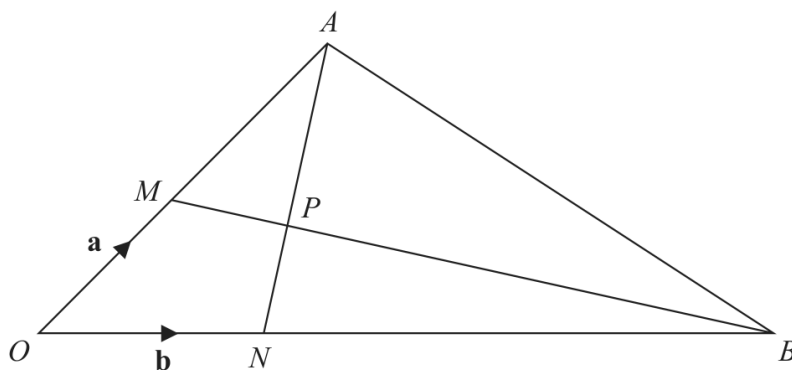
Using a vector method, find the ratio $AQ:QB$

$$AQ:QB = \dots\dots\dots$$

(Total for Question 21 is 5 marks)

Q38 **PAST PAPER** Edexcel 4MA1 • May 2023 • Paper 1H • Q19**CH 5 • TOPIC 1** **5 marks**

19

Diagram **NOT**
accurately drawn OMA, ONB, MPB and NPA are straight lines. M is the midpoint of OA $ON:NB = 1:5$

$$\vec{OM} = \mathbf{a} \quad \vec{ON} = \mathbf{b}$$

(a) Find in terms of $\mathbf{a}$ and $\mathbf{b}$ the vector $\vec{AN}$

.....

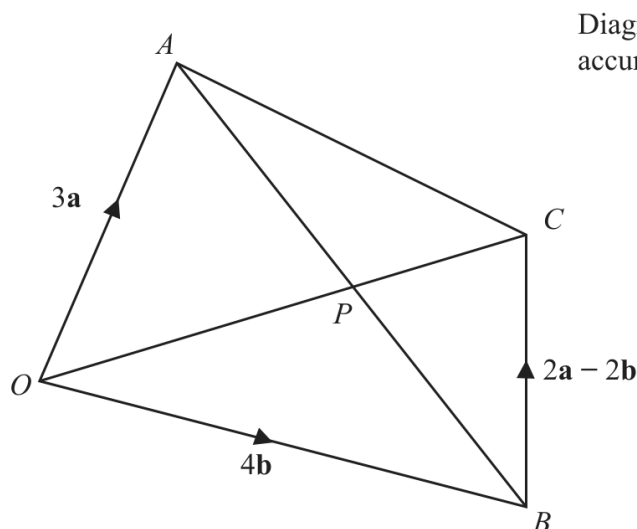
(b) Use a vector method to find the ratio $AP:PN$ $AP:PN =$ **(Total for Question 19 is 5 marks)**

Q39**PAST PAPER**

Edexcel 4MA1 • November 2023 • Paper 2H • Q22

CH 5 • TOPIC 1 5 marks

22



$OACB$ is a quadrilateral.

$$\vec{OA} = 3\mathbf{a} \quad \vec{OB} = 4\mathbf{b} \quad \vec{BC} = 2\mathbf{a} - 2\mathbf{b}$$

- (a) (i) Find the vector $\vec{OC}$ in terms of $\mathbf{a}$ and $\mathbf{b}$
Simplify your answer.

$$\vec{OC} = \dots\dots\dots$$

- (ii) Find the vector $\vec{AB}$ in terms of $\mathbf{a}$ and $\mathbf{b}$

$$\vec{AB} = \dots\dots\dots$$

The point P lies on AB and on OC

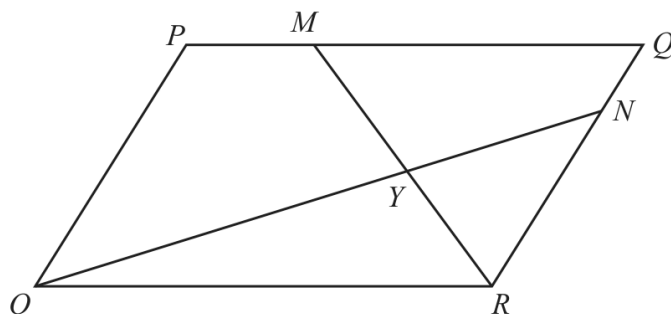
- (b) Using a vector method, find the ratio $AP : PB$
Show your working clearly.

.....

(Total for Question 22 is 5 marks)

Q40**PAST PAPER**

Edexcel 4MA1 • June 2024 • Paper 2HR • Q25

CH 5 • TOPIC 1 6 marks25 $OPQR$ is a parallelogram.Diagram **NOT**
accurately drawn

$$\vec{OP} = 2\mathbf{a} \quad \text{and} \quad \vec{OR} = 3\mathbf{b}$$

The point M lies on PQ such that $PM = \frac{1}{4}PQ$

The point N lies on RQ such that $RN = \frac{4}{5}RQ$

(a) Find, in terms of $\mathbf{a}$ and $\mathbf{b}$, giving your answers in simplest form

(i) $\vec{ON}$

.....

(ii) $\vec{MR}$

.....

MR and ON intersect at the point Y

Given that

$$OY = k \times ON$$

(b) use a vector method to find the value of k

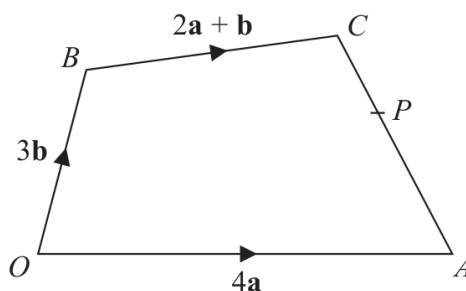
$$k = \dots\dots\dots$$

(Total for Question 25 is 6 marks)

Q41 PAST PAPER Edexcel 4MA1 • June 2024 • Paper 2H • Q24

CH 5 • TOPIC 1 6 marks

24

Diagram NOT
accurately drawn

The diagram shows a quadrilateral $OACB$ in which

$$\vec{OA} = 4\mathbf{a} \quad \vec{OB} = 3\mathbf{b} \quad \vec{BC} = 2\mathbf{a} + \mathbf{b}$$

- (a) Find $\vec{AC}$ in terms of $\mathbf{a}$ and $\mathbf{b}$
Give your answer in its simplest form.

$$\vec{AC} = \dots\dots\dots$$

The point P lies on AC such that $AP:PC = 3:2$

The point Q is such that OPQ and BCQ are straight lines.

- (b) Using a vector method, find $\vec{OQ}$ in terms of $\mathbf{a}$ and $\mathbf{b}$
Give your answer in its simplest form.
Show your working clearly.

$$\vec{OQ} = \dots\dots\dots$$

(Total for Question 24 is 6 marks)

Q42

PAST PAPER

Edexcel 4MA1 • November 2024 • Paper 1H • Q24

CH 5 • TOPIC 1

6 marks

24 OAB is a triangle.

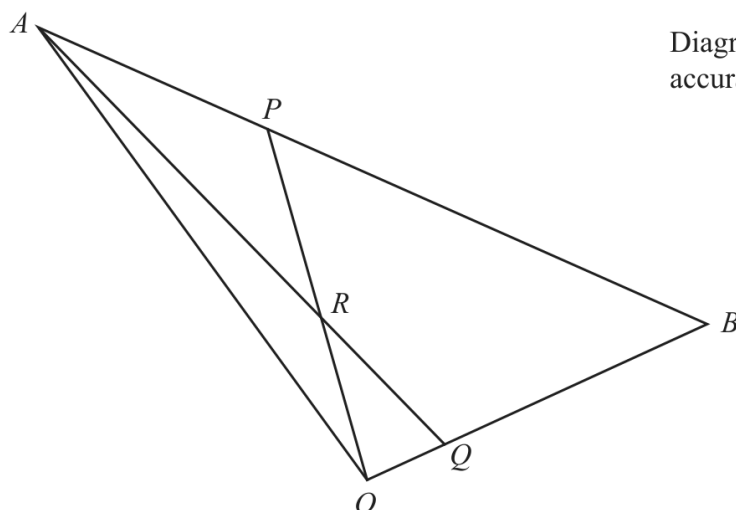


Diagram **NOT**
accurately drawn

$$\vec{OA} = 10\mathbf{a} \quad \vec{OB} = 10\mathbf{b}$$

ARQ and ORP are straight lines.

$$\vec{AP} = \frac{1}{4} \vec{AB} \quad \text{and} \quad \vec{OQ} = \frac{1}{5} \vec{OB}$$

Write the following vectors in terms of $\mathbf{a}$ and $\mathbf{b}$
Simplify your answers.

(i) $\vec{AQ}$

.....

(ii) $\vec{OP}$

.....

(iii) $\vec{OR}$

.....

(Total for Question 24 is 6 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|-----------------------|--|
| 01 | Column vectors | Number by number; magnitude by Pythagoras. |
| 02 | Paths | Known roads only; against the arrow → minus. |
| 03 | Ratios | $m : n \rightarrow \frac{m}{m+n}$ of the whole line. |
| 04 | Proof | Parallel: a multiple. Straight: a multiple and a common point. |
| 05 | Intersections | Two routes, s and t; match a and b . |

BOTTOM LINE Start → roads you know → finish. Parallel = a number times. Two routes meet = match coefficients.

Answers

Vectors • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • COLUMN VECTORS AND MAGNITUDE

- Q1** Dr Eslam • Magnitude $\begin{pmatrix} 7 \\ -2 \end{pmatrix}; \sqrt{53} = 7.28$
- Q2** Practice • Medium (a) arrow $\begin{pmatrix} -2 \\ -4 \end{pmatrix}$ (b) $\begin{pmatrix} 3 \\ -4 \end{pmatrix}$
- Q3** Practice • Medium $\begin{pmatrix} 3 \\ -2 \end{pmatrix}$
- Q4** Practice • Hard (a) A(3, 1), C(13, 12) (b) $\vec{AE} = \begin{pmatrix} 60 \\ 210 \end{pmatrix} = 30\begin{pmatrix} 2 \\ 7 \end{pmatrix}$: parallel with A in common
- Q5** Practice • Very Hard $2\mathbf{m} + \mathbf{n} = \begin{pmatrix} 7 \\ 2k+11 \end{pmatrix}; k = -9$
- Q6** Edexcel 4MA1 • January 2021 • Paper 1HR • Q13 $\begin{pmatrix} 3 \\ 7 \end{pmatrix}$
- Q7** Edexcel 4MA1 • May 2023 • Paper 1HR • Q19 $\sqrt{53} + \sqrt{34} - \sqrt{25} = 8.1 \text{ km}$
- Q8** Edexcel 4MA1 • November 2024 • Paper 2H • Q17 $\begin{pmatrix} 1 \\ -16 \end{pmatrix}; \sqrt{257} = 16.0$

IDEA 02 • VECTOR PATHS IN DIAGRAMS

Q9	Dr Eslam • Path	$\mathbf{b} - \mathbf{a}; \mathbf{a} - \mathbf{b}$
Q10	Practice • Medium	(a) $\mathbf{b} - \mathbf{a}$ (b) $\frac{1}{4}\mathbf{a} + \frac{3}{4}\mathbf{b}$
Q11	Practice • Medium	$3\mathbf{a} - 2\mathbf{b}$
Q12	Practice • Hard	$-\frac{1}{3}\mathbf{a}$
Q13	Practice • Very Hard	(a) $2\mathbf{a}$ (b) $n = 4$
Q14	Edexcel 4MA1 • November 2021 • Paper 1H • Q14	(a)(i) $\mathbf{b} - \mathbf{a}$ (ii) $4\mathbf{b} - 2\mathbf{a}$ (iii) $2\mathbf{b} - 3\mathbf{a}$ (b) $120 - 30 = 90 \text{ cm}^2$
Q15	Edexcel 4MA1 • June 2025 • Pa- per 2HR • Q18	(a) $4\mathbf{b} - 4\mathbf{a}$ (b) $3\mathbf{a} + \mathbf{b}$

IDEA 03 • RATIOS ON A LINE

Q16	Dr Eslam • Ratio	$6\mathbf{a} + \frac{1}{3}(6\mathbf{b} - 6\mathbf{a}) = 4\mathbf{a} + 2\mathbf{b}$
Q17	Practice • Hard	$\vec{OP} = \frac{3}{4}\mathbf{a} + \frac{5}{4}\mathbf{b} = \frac{1}{4}(3\mathbf{a} + 5\mathbf{b}); k = \frac{1}{4}$
Q18	Practice • Very Hard	$k = \frac{2}{5}$
Q19	Practice • Very Hard	$k = \frac{2}{5}$
Q20	Practice • Very Hard	$k = 4$
Q21	Edexcel 4MA1 • June 2022 • Pa- per 2HR • Q25	$\lambda = \frac{3}{7}, \mu = \frac{6}{7}$
Q22	Edexcel 4MA1 • June 2025 • Pa- per 1H • Q23	(a) $8\mathbf{b} - 12\mathbf{a}$ (b) $n = \frac{24}{5}$
Q23	Edexcel 4MA1 • November 2025 • Paper 1H • Q27	(a) $\mathbf{b} - 4\mathbf{a}$ (b) $n = \frac{32}{17}$

IDEA 04 • PROOF: PARALLEL AND COLLINEAR

- Q24** Dr Eslam • Parallel $6\mathbf{a} - 9\mathbf{b} = 3(2\mathbf{a} - 3\mathbf{b})$
- Q25** Practice • Medium $\vec{AB} = 2\mathbf{n} - 2\mathbf{m} = 2\vec{MN}$
- Q26** Practice • Hard (a) $3\mathbf{a} + 3\mathbf{b}$ (b) $\vec{AG} = 2\mathbf{b} - 4\mathbf{a}$, $\vec{AN} = 3\mathbf{b} - 6\mathbf{a} = \frac{3}{2}\vec{AG}$: straight line
- Q27** Practice • Hard $\vec{BC} = -2\mathbf{a} + 3\mathbf{b}$, $\vec{CD} = -10\mathbf{b}$: not multiples — No
- Q28** Practice • Very Hard (a) $\frac{1}{3}\mathbf{a} + \frac{2}{3}\mathbf{b}$ (b) $\vec{OD} = \mathbf{a} + 2\mathbf{b} = 3\vec{ON}$
- Q29** Practice • Very Hard $\vec{XY} = \frac{2}{3}\mathbf{a} + \mathbf{b} + \frac{2}{3}\mathbf{c}$, $\vec{PQ} = \mathbf{a} + \mathbf{b} + \mathbf{c}$: not a multiple — No
- Q30** Practice • Very Hard $\vec{EF} = \frac{1}{4}(3\mathbf{b} - \mathbf{a})$, $\vec{AG} = \frac{1}{2}(3\mathbf{b} - \mathbf{a}) = 2\vec{EF}$

IDEA 05 • INTERSECTIONS: TWO ROUTES TO ONE POINT

- Q31** Edexcel 4MA1 • January 2020 • Paper 2HR • Q26 $\vec{ON} = \frac{9}{5}\mathbf{a} + \frac{12}{5}\mathbf{b}$
- Q32** Edexcel 4MA1 • January 2020 • Paper 2H • Q23 $AM = 2\mathbf{a} + \mathbf{b}, AP = \frac{3}{4}AM: 3 : 1$
- Q33** Edexcel 4MA1 • January 2021 • Paper 1H • Q23 $4 : 1$
- Q34** Edexcel 4MA1 • May 2021 • Paper 1H • Q26 $\frac{5}{4}\mathbf{a} + \frac{15}{8}\mathbf{b}$
- Q35** Edexcel 4MA1 • January 2022 • Paper 1HR • Q24 $\lambda = \frac{3}{4}$
- Q36** Edexcel 4MA1 • January 2022 • Paper 1H • Q22 $\vec{OP} = 2\mathbf{a} + \frac{3}{2}\mathbf{b}$
- Q37** Edexcel 4MA1 • June 2022 • Paper 2H • Q21 $8 : 9$
- Q38** Edexcel 4MA1 • May 2023 • Paper 1H • Q19 (a) $\mathbf{b} - 2\mathbf{a}$ (b) $6 : 5$
- Q39** Edexcel 4MA1 • November 2023 • Paper 2H • Q22 (a)(i) $2\mathbf{a} + 2\mathbf{b}$ (ii) $4\mathbf{b} - 3\mathbf{a}$ (b) $3 : 4$
- Q40** Edexcel 4MA1 • June 2024 • Paper 2HR • Q25 (a)(i) $\frac{8}{5}\mathbf{a} + 3\mathbf{b}$ (ii) $-2\mathbf{a} + \frac{9}{4}\mathbf{b}$ (b) $k = \frac{5}{8}$
- Q41** Edexcel 4MA1 • June 2024 • Paper 2H • Q24 (a) $4\mathbf{b} - 2\mathbf{a}$ (b) $\frac{42}{5}\mathbf{a} + \frac{36}{5}\mathbf{b}$
- Q42** Edexcel 4MA1 • November 2024 • Paper 1H • Q24 (i) $2\mathbf{b} - 10\mathbf{a}$ (ii) $\frac{15}{2}\mathbf{a} + \frac{5}{2}\mathbf{b}$ (iii) $\frac{15}{4}\mathbf{a} + \frac{5}{4}\mathbf{b}$